CO1-AT3-TECHNICAL PRESENTATION

CSA0509 : Database Management Systems

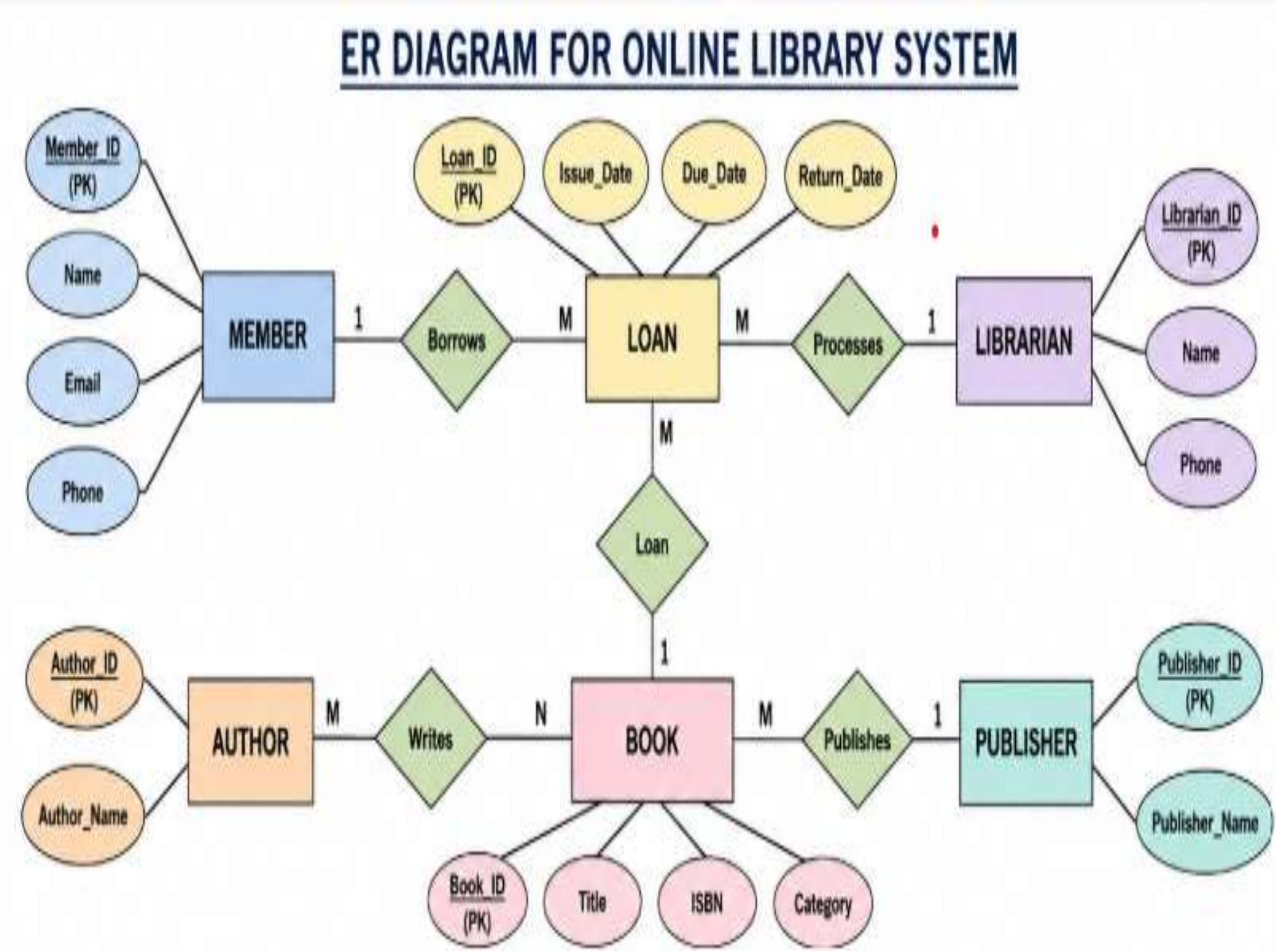
PRESENTED BY :
Shaik Ayesha Tabassum(192525074)

Q1. Create and present an ER diagram for an Online Library System, explaining each component clearly.

INTRODUCTION

- An **Online Library System** is a digital system used to manage library activities.
- It stores information about **books, members, authors, and librarians**.
- It helps users **search, borrow, and return books** easily.
- The **ER Diagram** shows the entities, attributes, and relationships in the library database.
- The system saves time, reduces paperwork, and makes library management simple and efficient.

ER DIAGRAM


ER DIAGRAM FOR ONLINE LIBRARY SYSTEM
Member_ID (PK)
Name
Email
Phone
MEMBER
1
Borrows
M
Loan_ID (PK)
Issue_Date
Due_Date
Return_Date
LOAN
M
Processes
1
LIBRARIAN
Librarian_ID (PK)
Name
Phone
M
Loan
1
Author_ID (PK)
Author_Name
AUTHOR
M
Writes
N
BOOK
M
Publishes
1
PUBLISHER
Publisher_ID (PK)
Publisher_Name
Book_ID (PK)
Title
ISBN
Category

COMPONENTS

Entities

- **Member** – Library user.
- **Book** – Book details.
- **Author** – Book writer.
- **Publisher** – Publishes books.
- **Loan** – Borrow and return records.
- **Librarian** – Manages library activities.

Relationships

Member → Loan: Borrows books (1:M)
Book → Loan: Loan records (1:M)
Author ↔ Book: Writes books (M:N)
Publisher → Book: Publishes books (1:M)
Librarian → Loan: Processes loans (1:M)

Attributes

Member: Member_ID, Name, Email, Phone
Book: Book_ID, Title, ISBN, Category
Author: Author_ID, Author_Name
Publisher: Publisher_ID, Publisher_Name
Loan: Loan_ID, Issue_Date, Due_Date, Return_Date
Librarian: Librarian_ID, Name, Phone

ADVANTAGES

- Easy management of books and members.
- Fast book issue and return process.
- Reduces paperwork and manual errors.
- Maintains accurate borrowing records.

CONCLUSION

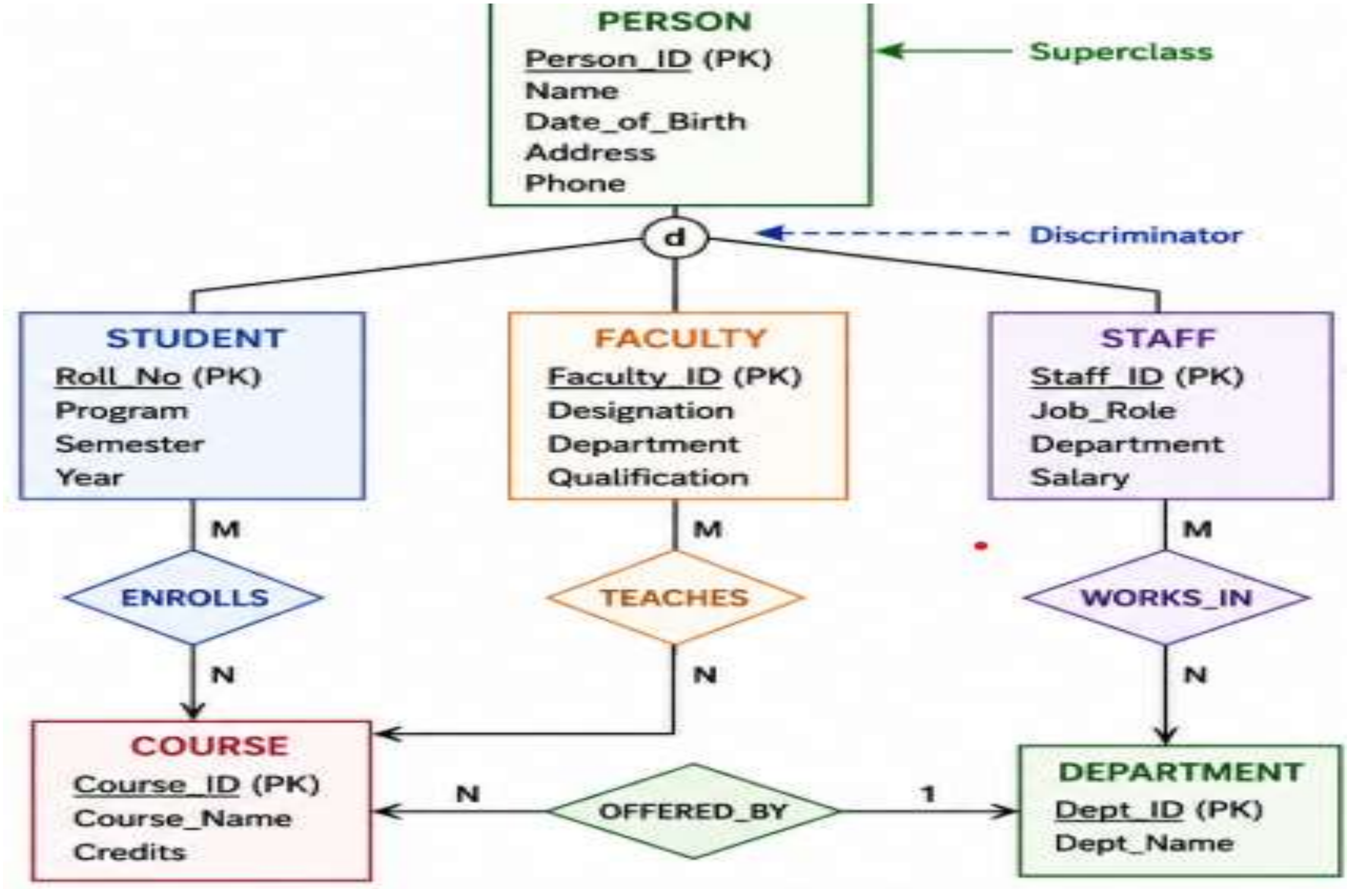
- The **Online Library System** makes library management easy and efficient.
- The **ER Diagram** shows the entities, attributes, and relationships clearly.
- It helps organize data accurately and reduces duplication.
- Overall, it provides a simple, reliable, and efficient database for library management.

Q2. Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.

INTRODUCTION

- The **Extended Entity-Relationship (EER) Model** is an advanced version of the ER model.
- It is used to design complex databases.
- The EER model includes **Generalization**, **Specialization**, and **Inheritance**.
- It helps organize data efficiently and reduces duplication.
- In a University system, **Person** is divided into **Student**, **Faculty**, and **Staff**.

UNIVERSITY CASE STUDY:



COMPONENTS

Superclass

- **Person** :Common entity that stores shared information**.**

Subclasses

- **Student** – Stores student details.
- **Faculty** – Stores faculty details.
- **Staff** – Stores staff details.

Generalization

- Combines similar entities into one **Superclass (Person)**.

Specialization

- Divides **Person** into **Student**, **Faculty**, and **Staff** based on their roles.

Inheritance

- Student, Faculty, and Staff inherit common attributes from **Person**

ADVANTAGES

- Reduces data duplication.
- Organizes data in a better way.
- Represents real-world situations clearly.
- Makes the database easy to manage.
- Improves database design and maintenance.

CONCLUSION

- The **EER Model** is an advanced version of the ER model.
- It uses **Generalization** and **Specialization** to organize data efficiently.
- It reduces data redundancy and improves database structure.
- In the University case study, **Person** is divided into **Student**, **Faculty**, and **Staff**.
- Overall, the EER model helps create a simple, efficient, and well-organized database.

Q3. Present the concept of data independence (logical and physical) and its significance in DBMS architecture.

INTRODUCTION

- **Data independence** is the ability to change the database structure without affecting application programs.
- It is an important feature of a **Database Management System (DBMS)**.
- It separates the data from the application programs.
- There are **two types of data independence**: Logical and Physical.
- It improves database flexibility, maintenance, and performance.

TYPES OF DATA INDEPENDENCE

Logical Data Independence

Ability to change the **logical structure** without affecting user applications.

Example:

Adding a new column (**Email**) to the **Student** table.

Existing programs continue to work without modification.

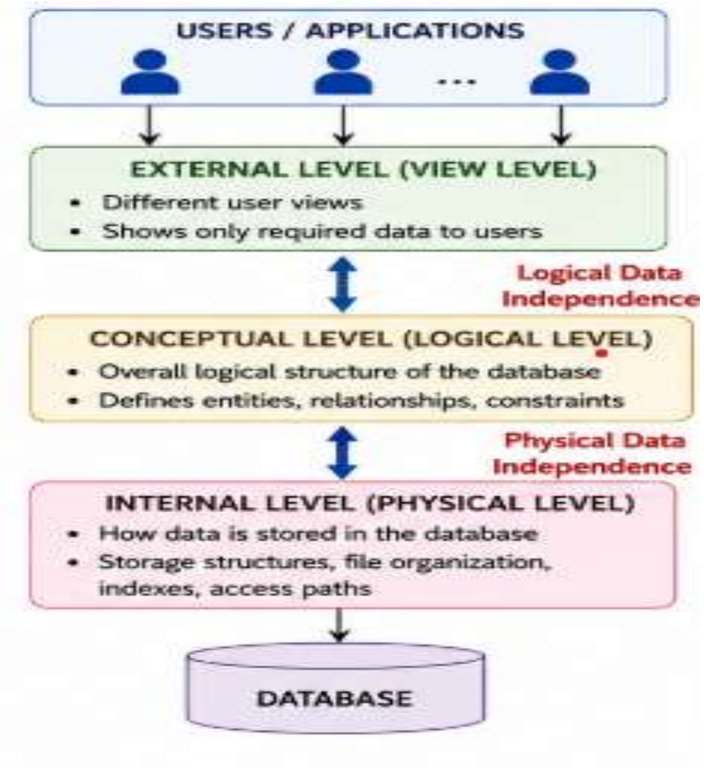
Physical Data Independence

Ability to change the **physical storage** without affecting the logical structure.

Example:

Creating an index to improve search speed.

Changing file organization or storage device.



ADVANTAGES & CONCLUSION

- Reduces maintenance cost.
- Improves database security.
- Increases flexibility.
- Allows easy database updates.
- Enhances system performance.

- Data independence is an important feature of DBMS.
- It separates applications from the database structure.
- Logical data independence protects applications from logical changes.
- Physical data independence protects applications from storage changes.
- Overall, it makes the database more flexible, efficient, and easy to maintain.

Q4. Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.

INTRODUCTION

- A **Database Management System (DBMS)** is software used to store, manage, and retrieve data.
- It helps users access data quickly and securely.
- A DBMS consists of different software components that work together.
- The main components are **Query Processor, Storage Manager, and Transaction Manager**.
- These components improve the performance, security, and reliability of the database.

COMPONENTS

Query Processor

- Accepts SQL queries from users.
- Checks the query for errors.
- Optimizes the query.
- Executes the query efficiently.

Transaction Manager

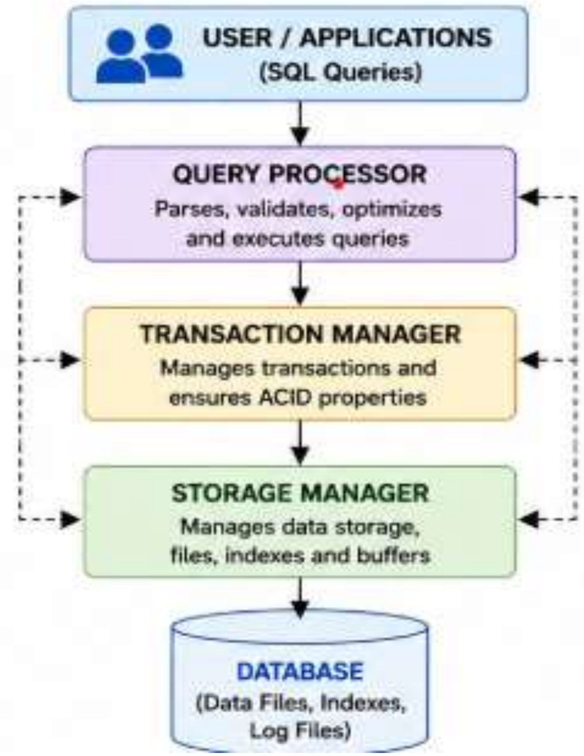
- Controls database transactions.
- Maintains **ACID** properties.
- Prevents data loss and inconsistency.
- Handles commit and rollback operations.

Storage Manager

- Stores and retrieves data from the database.
- Manages files, indexes, and memory.
- Organizes data for faster access.

WORK FLOW & DIAGRAM

- User sends an SQL query.
- Query Processor checks and executes the query.
- Transaction Manager manages the transaction safely.
- Storage Manager accesses the required data.
- The result is returned to the user.



CONCLUSION

- The **Query Processor** processes SQL queries.
- The **Storage Manager** manages data storage and retrieval.
- The **Transaction Manager** ensures safe and reliable transactions.
- All three components work together for efficient database management.
- They make the DBMS fast, secure, and reliable.

Q5.Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.

INTRODUCTION

- An **Entity** is an object or thing that stores data in a database.
- There are **two types of entities**: Strong Entity and Weak Entity.
- A **Strong Entity** has its own primary key.
- A **Weak Entity** depends on a strong entity for identification.
- ER modeling uses both entities to represent real-world data accurately**.**

Comparison Table

Strong Entity	Weak Entity
Has its own Primary Key	Does not have its own Primary Key
Exists independently	Depends on a Strong Entity
Represented by a single rectangle	Represented by a double rectangle
Primary key uniquely identifies records	Uses a partial key with the owner's key
Example: Student, Employee	Example: Dependent, Order Item

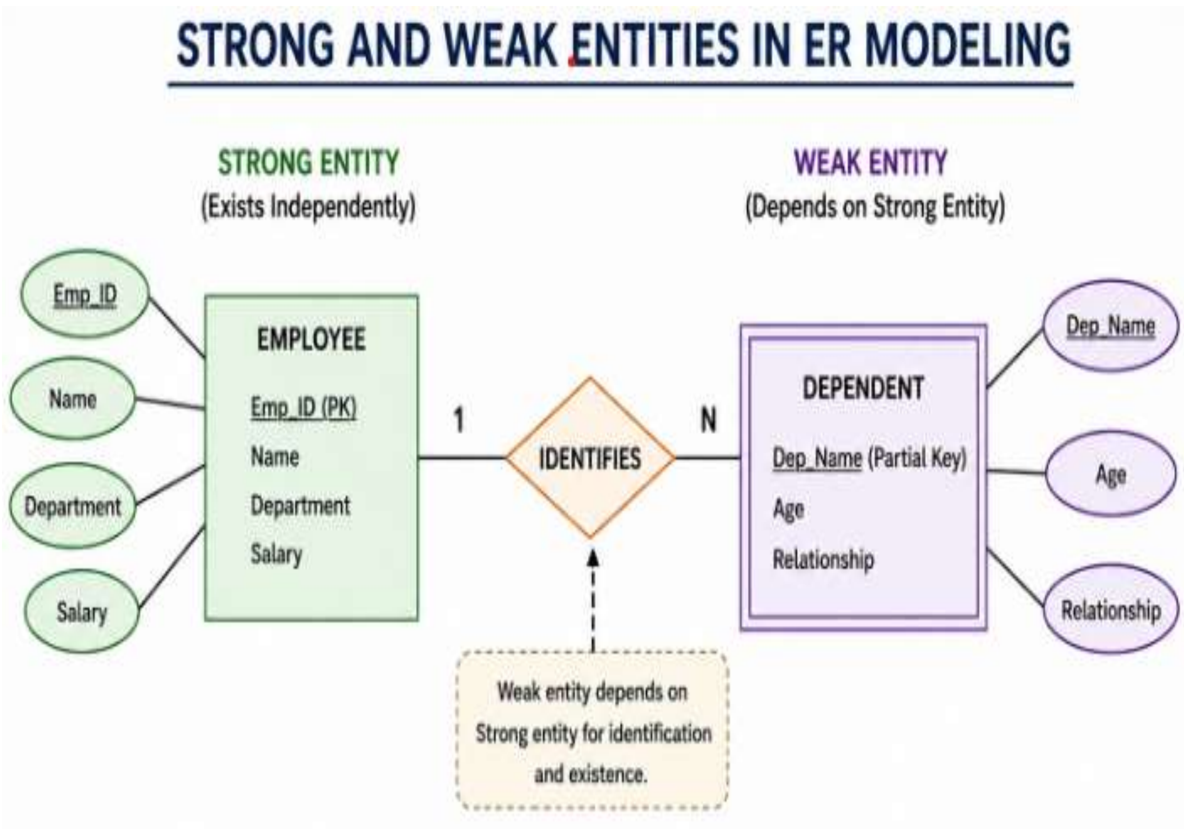
COMPONENTS

Strong Entity
- Exists independently.
- Has its own Primary Key.
- Example: Student, Employee, Book.

Weak Entity
- Depends on a Strong Entity.
- Does not have a complete Primary Key.
- Example: Dependent, Order Item.

Relationship
- Strong Entity identifies the Weak Entity.
- The Weak Entity cannot exist without the Strong Entity.

STRONG & WEAK ENTITIES



CONCLUSION

- A **Strong Entity** has its own primary key and exists independently.
- A **Weak Entity** depends on a strong entity for identification.
- Both entities help represent real-world scenarios correctly.
- They improve the accuracy and organization of database design.
- ER modeling makes databases simple, efficient, and reliable.